Zixiang Xu

CONTACT INFORMATION

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Ph.D. Student
Computer Science
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RESEARCH SUMMARY

I study how to evaluate and improve foundation models and agents, and how to determine when their outputs and decisions can be trusted. My work connects **evaluation and post-training** with **trustworthy AI** through four directions:

1. Frontier Model Evaluation: Capability Boundaries and Interactive Tasks

A benchmark score is useful only if it tells us what a model can actually do. I build tasks and executable environments that test reasoning over graphs, changing worlds, social interactions, and tool-based optimization. For open-ended mathematical research, I develop expert evaluation of problem posing, judging, and development. Across these settings, I study how task structure and evaluation protocols can expose capability boundaries that aggregate scores miss.

- ❑ Benchmark Design
- ❑ Reasoning & World Modeling
- ❑ Interactive Environments
- ❑ Tool Use & Optimization
- ❑ Expert Evaluation
- ❑ Mathematical Research

2. Evaluation-Driven Post-Training: Data, Environments, and Feedback

Failures become useful for learning when they can be turned into dependable supervision. I develop environments that generate verifiable agent trajectories, curate reasoning traces for post-training, and use automatically discovered cross-lingual failures for targeted training. The central question is whether learning from a failure teaches a reusable capability or only repairs the tested examples. I therefore evaluate transfer beyond the data and language used for training.

- ❑ Post-Training
- ❑ RLVR Environments
- ❑ Agent Trajectories
- ❑ Reasoning Traces
- ❑ Failure-Driven Training
- ❑ Cross-Lingual Transfer

3. Automated Failure Discovery: Language, Context, and Decisions

Strong average performance can coexist with predictable failures on particular inputs. I develop search-based probes that find multilingual and contextual weaknesses, including cases where a change in language or surrounding text alters a decision without changing the underlying task. Controlled perturbations help distinguish capability limitations from sensitivity to incidental information. This work actively searches for failure conditions that a fixed test distribution may rarely encounter.

- ❑ Automated Probing
- ❑ Adversarial Search
- ❑ Cross-Lingual Reliability
- ❑ Context Sensitivity
- ❑ Decision-Model Reliability
- ❑ Counterfactual Evaluation

4. Evaluator Reliability and Agent Risk: Mechanisms and Verification

An evaluator can be wrong even when the answer it scores is right; using that score for training can reinforce the error. I study how LLM-judge bias is represented internally, how causal interventions change scoring, and how those representations predict evaluator failures. I also build automated tests for rule-based verifiers and study risk when agents, tools, or workflows change. The shared question is what evidence justifies trusting a score, a reward, or an agent's decision.

- ❑ LLM-as-a-Judge
- ❑ Mechanistic Interpretability
- ❑ Causal Interventions
- ❑ Verifier & Reward Reliability
- ❑ Failure Prediction
- ❑ Agent Risk Assessment

Education

University of Southern California (USC) — *Los Angeles, CA*
Ph.D. Student in Computer Science · Advisor: Prof. Yue Zhao — Aug. 2026 – Present
FORTIS Lab. Expected graduation: 2030.

Sichuan University — *Chengdu, China*
B.Eng. in Computer Science and Technology — Sep. 2022 – Jun. 2026

Research Experience

Meta AI — *Remote*
Research Collaboration — Sep. 2026 – Present
Lead Grid-Bench: executable spatial environments for frontier-model evaluation and post-training.

Tencent (WeChat AI) — *Beijing, China*
Research Intern · Foundation Model Group — Jun. 2026 – Aug. 2026
Contributed to WeLM post-training. Led AlgoWorlds, building 240 verifiable tool-use worlds across ten optimization families and generating agent trajectories for post-training.

Alibaba Group — *Beijing, China*
Research Intern — Mar. 2026 – May 2026
Led Inside the Unfair Judge, studying scoring-bias mechanisms, causal bias control, and failure prediction to inform the reliability of automated feedback for post-training.

MBZUAI — *Abu Dhabi, UAE*
Research Intern · Supervisor: Prof. Xiuying Chen — Mar. 2025 – Aug. 2025
Led SocialMaze, GTA, and DyFlow: social-reasoning post-training with curated traces, algorithmic graph reasoning, and dynamic agent workflows.

MINE Lab, University of Notre Dame — *Remote*
Research Intern · Supervisor: Prof. Xiangliang Zhang — Oct. 2024 – Aug. 2025
Led Cross-Lingual Pitfalls and Adaptive Distraction, automating failure discovery and using cross-lingual failures for targeted post-training and transfer evaluation.

MIT Center for Transportation & Logistics — *Massachusetts, USA*
Summer Research Student · Supervisor: Prof. Elenna R. Dugundji — Aug. 2024
Led Temporal-IRL, applying inverse reinforcement learning to port congestion and berth scheduling; selected for a TRB 2025 oral presentation.

Microsoft (Asia) Internet Engineering Institute — *Beijing, China*
Research Intern · Supervisor: Wang Pan — Jan. 2024 – Mar. 2024
Led LLM and VLM evaluation, including GPT-4V capability assessment and reproduction.

Honors and Awards

USC Annenberg Fellowship — 2026
Competitive fellowship top-off for selected USC Viterbi Ph.D. students.

CCF Elite Collegiate Award — 2024
One of 100 junior and senior undergraduates selected nationwide; recognized for academics, research, and service. Invited speaker/session host at CNCC; team placed **3rd nationwide** in the CCSP Team Contest.

Gold Medal, The 2022 ICPC Asia Jinan Regional Contest — 2022
Gold medal in the ICPC, the world's premier collegiate algorithmic programming competition.

Silver Medal, National Olympiad in Informatics (NOI) — 2020
Silver medal in China's national informatics olympiad for secondary-school students, organized by the China Computer Federation (CCF).

Research Software & Platforms

RewardAudit: Automated Testing of Rule-Based Verifiers for LLM Evaluation and RLVR

MathInsight Arena: Expert Evaluation of LLMs on Posing, Judging, and Developing Mathematical Research Problems

Publications

1. **Zixiang Xu**, Zirui Song, Wanrong Zheng, Chiyu Zhang, Yue Huang, Yuyang Peng, Kexin Zhang, Jinkun Chen, Xiuying Chen, Xiangliang Zhang, Yan Liu, Laurent Itti, Frank Shyu, Xiaoqiang Luo, Sandeep Pandey, Xi Liu, Luke Simon, Xiyang Hu, Yue Zhao.
   Grid-Bench: Can LLMs Predict How Discrete Worlds Change?
   Preprint, 2026. [Project] [Paper] [Code] [Data]

2. **Zixiang Xu**, Jiaan Wang, Fandong Meng.
   AlgoWorlds: Benchmarking Tool Use for Global Optimization in Algorithmic Worlds
   Preprint, 2026. [Project] [Paper] [Code & Data]

3. Yue Zhao, **Zixiang Xu**, Angela Huyue Zhang, S. Alex Yang, Xiyang Hu.
   agent-risk: Risk Assessment for Changes to AI Agents
   Preprint, 2026. [Paper] [Code & Data]

4. **Zixiang Xu**, Yanbo Wang, Yue Huang, Haomin Zhuang, Yujun Zhou, Jiayi Ye, et al.
   SocialMaze: A Benchmark for Evaluating and Enhancing Social Reasoning in Large Language Models in Complex Social Environments
   Findings of EMNLP 2026; previously SocialSim @ **COLM 2025** Workshop (Spotlight).
   [Project] [Paper] [Code] [Data]

5. **Zixiang Xu**[*], Yanbo Wang[*], Yue Huang[*], Xiuying Chen, Jieyu Zhao, Meng Jiang, Xiangliang Zhang ([*]Equal contribution).
   Cross-Lingual Pitfalls: Automatic Probing Cross-Lingual Weakness of Multilingual Large Language Models
   ACL 2025. [Project] [Paper] [Code]

6. Yanbo Wang[*], **Zixiang Xu**[*], Yue Huang[*], Chujie Gao, Siyuan Wu, Jiayi Ye, et al. ([*]Equal contribution).
   Adaptive Distraction: Probing LLM Contextual Robustness with Automated Tree Search
   NeurIPS 2025. [Paper] [Code]

7. Yanbo Wang, **Zixiang Xu**, Yue Huang, Xiangqi Wang, Zirui Song, Lang Gao, Chenxi Wang, et al.
   DyFlow: Dynamic Workflow Framework for Agentic Reasoning
   NeurIPS 2025. [Paper] [Code]

8. **Zixiang Xu**, Zirui Song, Chiyu Zhang, Xiuying Chen, Xi Liu, Xiyang Hu, Yue Zhao.
   JevOut: Natural Context Can Flip Decision Models
   Preprint, 2026. **Agenthon @ NeurIPS 2026** (Poster).
   [Project] [Paper] [Code & Data]

9. **Zixiang Xu**, Sixian Li, Huaxing Liu, Xiang Wang, Shuai Li, Zirui Song, Xiuying Chen.
   Inside the Unfair Judge: A Mechanistic Interpretability Account of LLM-as-Judge Bias
   Preprint, 2026. [Project] [Paper]

10. **Zixiang Xu**, Yanbo Wang, Chenxi Wang, Lang Gao, Zirui Song, Yue Huang, Zhaorun Chen, Xiangliang Zhang, Xiuying Chen.
    GTA: Graph Theory Agent and Benchmark for Algorithmic Graph Reasoning with LLMs
    Preprint, 2026. [Project] [Paper] [Code & Data]

Publications (continued)

11. Chenxi Wang, Yixuan Zhang, et al., **Zixiang Xu**, et al.
Do LLMs "Feel"? Emotion Circuits Discovery and Control
ICML 2026. [Paper] [Code]

12. Zirui Song, Qian Jiang, et al., **Zixiang Xu**, et al.
Audio Jailbreak: An Open Comprehensive Benchmark for Jailbreaking Large Audio-Language Models
ACL 2026. [Paper] [Code]

13. Zirui Song, Guangxian Ouyang, et al., **Zixiang Xu**, et al.
ManipLVM-R1: Reinforcement Learning for Reasoning in Embodied Manipulation with Large Vision-Language Models
AAAI 2026. [Paper]

14. Yikuan Hu[*], **Zixiang Xu**[*], Wei Zhang[*], et al. ([*]Equal contribution).
Temporal-IRL: Modeling Port Congestion and Berth Scheduling with Inverse Reinforcement Learning
TRB 2025 (Oral). [Paper]

15. Chenxi Wang, Yixuan Zhang, Lang Gao, **Zixiang Xu**, et al.
Under the Shadow of Babel: How Language Shapes Reasoning in LLMs
Findings of EMNLP 2025. [Paper]

16. Ruowei Wang, Jiaqi Li, et al., **Zixiang Xu**, et al.
GenUDC: High-Quality 3D Mesh Generation with Unsigned Dual Contouring Representation
ACM MM 2024. [Paper]

17. Zirui Song, Yuan Huang, Junchang Liu, et al., **Zixiang Xu**, et al.
Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies
EMNLP 2026. [Paper]

18. Lang Gao, Kaiyang Wan, Wei Liu, et al., **Zixiang Xu**, et al.
Evaluate Bias without Manual Test Sets: A Concept Representation Perspective for LLMs
Preprint, 2025. [Paper]

19. Yu Liu, Ruowei Wang, Jiaqi Li, **Zixiang Xu**, Qijun Zhao.
MTFusion: Reconstructing Any 3D Object from Single Image Using Multi-word Textual Inversion
PRCV 2024 (Oral). [Paper]

20. Yuchuan Deng, Qijun Zhao, Zhanpeng Hu, **Zixiang Xu**.
Text-Based Face Retrieval: Methods and Challenges
CCBR 2023 (Oral). [Paper]

Academic Service

Conference Reviewer: AAAI 2027; ICLR 2026–2027; COLM 2026; NeurIPS 2026; NLPCC 2025.

ACL Rolling Review Reviewer: February and May 2025; August and October 2026 cycles.

Workshop Reviewer: VLM4RWD at NeurIPS 2026; GroundLM and REALM at EMNLP 2026; Social Simulation with LLMs at COLM 2025.